Referee report on "On the transfer of Shalika germs for G_2" by David Joyner (manuscript, summer 1994)

Referee: Claude (Anthropic), an AI system

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Abstract

This report was written by an AI system (Claude, made by Anthropic) at the author's request. It is not an independent certification of the results. The manuscript proposes to prove the matching of the 1-, 2- and 3-regular Shalika germs of the split exceptional group G_2 with germs on its cuspidal endoscopic groups, and hence local transfer for G_2. The geometric part of the paper is largely sound. I verified by computer that the variety Z of §3 is normal, that the essential patches of §4 are regular and cover Z up to codimension two, and that the Igusa constants of the 1-, 2- and 3-regular base divisors are 2, 3 and 4, as claimed. However, the proofs of the main theorems (Theorems 6.1, 7.2 and 8.3) are not complete. They rest on Hales' reduction lemma (Lemma 5.19), whose hypothesis cannot be met for the cuspidal endoscopic groups of G_2. The hypotheses on the support of f in those theorems can never be satisfied. The endoscopic group $PGL(3)$ is given the wrong roots. Several elliptic tori that carry nontrivial κ are missing. There are also many misprints, notational clashes and unproved assertions. I recommend a major revision before the manuscript is circulated as a proof of transfer for G_2.

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1 What I read, and how I checked it

Files read. I read all fifteen uploaded files completely: `g2new0.tex` through `g2new9.tex`, the variants `g2new5d.tex` and `g2new9d.tex`, `g2newapp.tex`, `g2newref.tex`, and the house style guide `joyner-math-paper-style-guide.md`. I also read the text that follows `\enddocument` in `g2new2`, `g2new4`, `g2new6` and `g2newapp`. That text was never typeset, but it bears on some issues below. I refer to it as "untypeset text".

Sources consulted. I could not consult Hales' thesis [H1], Hales' $U(3)$ paper [H3], Langlands [L], Langlands–Shelstad [LS0]–[LS3], Kempf et al. [K], Spaltenstein [Spal] or any other cited source in full. I checked bibliographic data against online records (EuDML, Numdam, the IAS archive of Langlands' papers, and a reference list in an AMS journal). I consulted Hales' paper [H2] (Astérisque 171–172) only through a machine-generated summary of its Numdam PDF. Where I rely on that summary I say so.

Verification levels. Following §7 of the style guide, every point carries a level. L1: an independent derivation by me, by hand or by an exact computation whose script is attached. L2: a check against a source I actually consulted. L3: reliance on memory of the literature. Every issue also carries a verdict: *correct*, *misprint*, *fixable gap* or *false*.

Computations. The following Python scripts accompany this report. They use exact integer and symbolic arithmetic (SymPy), and they run in a few minutes.

- `g2_toric_divisors.py`: the toric structure of Z, its facets, all branches of $\lambda = 0$ with $a(E)$, $b(E)$, $\beta(E)$, and which patches meet which divisors.
- `g2_saturation_full.py`: a complete proof that the semigroup of Z is saturated, so that Z is normal.
- `g2_patch_jacobians.py` and `g2_patch_regularity.py`: the patch formulas of §3, the Igusa forms of §4, and the forms (8.1) and (8.4).
- `g2_exceptional_divisors.py`: the singular torus orbits of Z and candidate exceptional divisors of a resolution.
- `g2_tori_endoscopy.py`: $H^1(F,T)$ for every elliptic torus of split G_2, and the root system of the endoscopic group for every κ.

2 Summary of the manuscript

Section 0 states the goal. The goal is to prove the matching of Shalika germs for the r-regular unipotent orbits of G_2, $r = 1, 2, 3$, by Langlands' Igusa method. Together with Shelstad's result for $r = 0$ and Rogawski's for the trivial orbit, this is said to give the smooth matching (local transfer) theorem for G_2. Section 1 recalls the variety of stars, Langlands' coordinates $z(W, \rho)$, the partial resolution S_1 and the varieties Y'', Y_1 and Y_C. Section 2 computes the Galois action on coordinates. Section 3 introduces the monomial variety Z and the "fundamental" and "essential" patches, and claims that Z is normal and Cohen–Macaulay. Section 4 lists the patches for G_2 and computes Igusa constants. Section 5 recalls Igusa's expansion, proves a "no spurious divisors" lemma and states Hales' reduction lemma. Sections 6, 7 and 8 treat the 1-, 2- and 3-regular germs, with endoscopic groups $PGL(3)$ and $(SL(2) \times SL(2))/\{\pm 1\}$. Section 9 states a general conjecture. The appendix computes a Chevalley basis via triality, the cuspidal endoscopic groups, $H^1(\Gamma, T)$ for some tori, and the stability of the 2- and 3-regular orbits.

The strategy has three steps. First, identify the unique base divisor that contributes to each germ. Second, show that its Igusa coefficient depends on the tangent direction X only through a product of roots of H. Third, apply Hales' reduction lemma to avoid computing that dependence exactly. The third step does not work for G_2 (Issue 4.1), and the second is asserted rather than computed (Issue 4.6).

3 Overall assessment

The manuscript contains real and useful work. In particular, it gives an explicit treatment of the Igusa variety of G_2, the essential patches, and the identification of the base divisors $E_{\emptyset,\alpha,\delta}$ and $E_{\emptyset,\beta,\zeta}$ that carry the 2- and 3-regular germs. My independent computations confirm the core geometric numbers (Section 6).

The main theorems are not established, for four independent reasons.

1. The descent step, Lemma 5.19, cannot be applied to the cuspidal endoscopic groups of G_2 (Issue 4.1).
2. The hypotheses of Theorems 6.1, 7.2 and 8.3 are never satisfied (Issue 4.2).
3. The roots of the endoscopic group $PGL(3)$ are misidentified (Issue 4.3).
4. Some elliptic endoscopic data are missing, so the smooth matching theorem would not follow even if the three theorems were proved (Issue 4.4).

The claim in §0 that the paper "proves the smooth matching theorem for G_2" should be withdrawn or made conditional.

For context in a revision (L3): local transfer for p-adic groups is now known in general. Waldspurger showed that the fundamental lemma implies transfer ("Le lemme fondamental implique le transfert", Compositio Math. 105 (1997)), and Ngô proved the fundamental lemma (Publ. Math. IHÉS 111 (2010)). A revision should therefore present the paper as an explicit Igusa-theoretic computation of the G_2 germs, not as the only route to transfer.

4 Major issues

Issue 4.1 (Hales' reduction lemma does not apply)**. Where:** §0 (the paragraph on $\Delta_0(X)$), Lemma 5.19, and the ends of the proofs of Theorems 6.1, 7.2 and 8.3. Lemma 5.19 needs Y in the Lie algebra of the center of the endoscopic group H with $\mathrm{Cent}(Y,G) = H$. Both cuspidal endoscopic groups of G_2, namely $PGL(3)$ and $(SL(2)\times SL(2))/\{\pm 1\}$, are semisimple of rank two, the rank of G_2. So $\mathrm{Lie}(Z(H)) = 0$. Moreover, the centralizer of a semisimple element Y of $\mathfrak{t}$ is a Levi subgroup. The Levi subgroups of G_2 are T, the two $GL(2)$'s and G_2. None of them is $SL(3)$ or $SO(4)$, which are the subgroups of G_2 the paper uses as "$H \subset G$". So no admissible Y exists. Read literally, the hypothesis is vacuous, and the lemma would give transfer with no information about the germs at all. That cannot be the intended statement.

The machine summary of [H2] that I consulted supports this reading. Hales' quasi-split reduction (Prop. 2.11 of [H2]) descends to elements s of the connected center torus S of H close to the identity. For the cuspidal $SO(4)$ case of $GSp(4)$, Hales instead computes the dependence on the tangent direction explicitly: the germ is $|T_1 - T_2|\, c(T,\kappa)$ (§6 of [H2]). The G_2 case needs the same kind of explicit computation, which §0 says the descent "remarkably" avoids.

Verdict: *false as applied; the proofs of Theorems 6.1, 7.2 and 8.3 are incomplete*. **Level:** L1 for the Lie-theoretic facts; L2 (machine summary only) for [H2]; L3 for [H3, Lemma 6.1], which I could not consult.

Suggested repair: Compute the X-dependence of the full coefficient $\kappa(t_*)\,|\omega_{E(u)}|$, including the κ-factor, as in [H2, §6]. Then compare it with the H-side directly.

Issue 4.2 (The support hypotheses are never satisfied)**. Where:** Theorems 6.1, 7.2 and 8.3; the decomposition $f = f_1 + f_2$ before Lemma 5.19; the conclusion of Lemma 5.19; Conjecture 9.1; the text of §§6–8. Each theorem assumes that the support of f meets the r-regular orbit $\mathcal{O}_r$ and no other unipotent orbit. The support of $f \in C_c^\infty(G(F))$ is open. Every unipotent $u \in G(F)$ lies in the unipotent radical N of an F-Borel subgroup (Borel–Tits, G_2 being split). Regular unipotents form a Zariski-dense open subset of $N \cong \mathbb{A}^6$, so they are p-adically dense in $N(F)$. Hence every neighborhood of u meets the regular orbits. An f whose support meets $\mathcal{O}_r$ also meets $\mathcal{O}_0$ and every orbit between them. The hypotheses are therefore never satisfied, and the theorems are vacuous as stated. The same objection applies to f_2 in the decomposition before Lemma 5.19. The phrase "orbits outside the Zariski closure of $\mathcal{O}$" names the *larger* orbits, which f_2 cannot avoid.

Verdict: *false as stated; fixable*. **Level:** L1.

Suggested repair: State the theorems as germ identities. Alternatively, use test functions with $\mu_{\mathcal{O}'}(f) = 0$ for every unipotent class $\mathcal{O}' \neq \mathcal{O}$. Such functions exist because the unipotent orbital integrals are linearly independent.

Issue 4.3 (The roots of the endoscopic group $PGL(3)$)**. Where:** §8 (the definition of $\Sigma^+_{H_1}$, (8.4) and (8.7)); §4, Lemma 4.4; Appendix A.5; untypeset text after §6, (6.8). The paper takes $\Sigma^+_{H_1} = \{\beta, \varepsilon, \zeta\}$, the long roots. That is the root system of the *pseudo-Levi subgroup* $SL(3) \subset G_2$. The *endoscopic group* $H_1 = PGL(3)$ is different. Its roots, transported to T_G, are the roots a of G with $\kappa(a^\vee) = 1$. For the cubic torus (Galois image $\langle (s_\beta s_\alpha)^2 \rangle$) we have $H^1(F,T) = X_*/(\sigma - 1)X_* \cong \mathbb{Z}/3$. In the coroot basis $(\alpha^\vee, \beta^\vee)$, $(\sigma - 1)X_*$ is spanned by $(1,0)$ and $(0,3)$. A nontrivial κ is trivial exactly on the coroots $\alpha^\vee = (1,0)$, $\gamma^\vee = (1,3)$ and $\delta^\vee = (2,3)$. So

$$\Sigma^+_{H_1} = \{\alpha,\ \alpha+\beta,\ 2\alpha+\beta\},$$

the *short* roots. Equivalently, $\widehat{H}_1 = SL(3)$ is generated by the long roots of $\widehat{G} = G_2(\mathbb{C})$, which are the coroots of the short roots of G. The set $\Delta_2^\vee = \{\beta^\vee, (3\alpha+\beta)^\vee\}$ used in A.5 consists of short roots of $\widehat{G}$, and they do not span a closed subsystem: $\beta^\vee + \zeta^\vee = \gamma^\vee$ is a root.

For H_2 the paper's $\Sigma^+_{H_2} = \{\alpha, \zeta\}$ is correct, because there the pseudo-Levi and the endoscopic root systems agree.

A short root system can still parametrize $\mathcal{O}_3$ through the long-root pseudo-Levi, so the coordinates in §8 need not change. But (8.7), and the claim after it that $p_{\mathcal{O}_3,T}$ is a product over roots "in H", must use the short roots, because D_{H_1} and the transfer factor do.

Verdict: *false; fixable*. **Level:** L1 (script `g2_tori_endoscopy.py`).

Issue 4.4 (Missing elliptic endoscopic data)**. Where:** Appendix A.6 ("All other non-split Cartans ... do not contribute"), Theorems 6.1, 7.2 and 8.3, the proof of Lemma 5.17, and §0. I computed $H^1(F,T) = \hat{H}^{-1}(\Gamma, X_*)$ for every elliptic Galois image $\Gamma \subset W(G_2)$:

Galois image in W	$H^1(F,T)$	roots of H for $\kappa \neq 1$
$\{\pm 1\}$ (quadratic T)	$(\mathbb{Z}/2)^2$	$\{\alpha,\zeta\}$, $\{\beta,\delta\}$, $\{\gamma,\varepsilon\}$
$\langle s_\alpha, s_\zeta\rangle$ and conjugates (biquadratic)	$\mathbb{Z}/2$	one orthogonal pair
cyclic of order 3	$\mathbb{Z}/3$	$\{\alpha,\gamma,\delta\}$
S_3 containing the short reflections	$\mathbb{Z}/3$	$\{\alpha,\gamma,\delta\}$
cyclic of order 6; S_3 with long reflections; W	0	—

The quadratic tori and the short-reflection S_3 tori carry nontrivial κ and are omitted. The proof of Lemma 5.17 also asserts that $H^1(\Gamma, T_G)$ is cyclic of order $m \in \{2,3\}$, which fails for quadratic tori.

The conclusions $H^1 = \mathbb{Z}/2$, 0 and $\mathbb{Z}/3$ in A.6 are right, but the computations given for them are not.

- The displayed formula $s_\beta s_\alpha(m\alpha^\vee + n\beta^\vee) = (2m-n)\alpha^\vee + (m-n)\beta^\vee$ has determinant -1, so it cannot be a rotation. The correct formula is $(n-m)\alpha^\vee + (2n-3m)\beta^\vee$.
- The "cubic" formula has determinant 3.
- The two coboundary identities are wrong.
- The biquadratic computation takes a quotient of H^1 groups, which is not valid. The correct computation is $(s_\alpha - 1)X_* + (s_\zeta - 1)X_* = \mathbb{Z}\alpha^\vee + \mathbb{Z}\zeta^\vee$, of index 2.

Verdict: *false (the claim of completeness); misprints in the computations*. **Level:** L1 (script).

Issue 4.5 (Which divisors contribute)**. Where:** §4 (the table (4.3) and the dismissal of exceptional divisors), §6 ("three base divisors with $\beta = 2$"), Lemma 0.8, Lemma 5.14, §§7–8. The toric computation lists every branch of $\lambda = 0$ on Z:

branch	a	b	β	image	patches meeting it
$E_{\{\alpha,\beta\}}$	1	1	1	regular	$Z_{\{\alpha,\beta\}}$
$E_\alpha = E_{\emptyset,\alpha,\gamma}$	1	2	2	subregular	Z_α, $Z_{\emptyset,\alpha,\gamma}$
$E_\beta = E_{\emptyset,\beta,\gamma} = E_{\emptyset,\beta,\delta} = E_{\emptyset,\beta,\varepsilon}$	1	2	2	subregular	Z_β, $Z_{\emptyset,\beta,\gamma/\delta/\varepsilon}$
$E_{\emptyset,\alpha,\delta}$	2	6	3	2-regular	$Z_{\emptyset,\alpha,\delta}$
$E_{\emptyset,\alpha,\varepsilon} = E_{\emptyset,\alpha,\zeta}$	3	11	$11/3$	—	$Z_{\emptyset,\alpha,\varepsilon}$, $Z_{\emptyset,\alpha,\zeta}$
$E_{\emptyset,\beta,\zeta}$	2	8	4	3-regular	$Z_{\emptyset,\beta,\zeta}$
$E_\emptyset$	1	7	7	trivial	$Z_\emptyset$

Four points follow.

(a) $E_{\emptyset,\alpha,\gamma}$ is the same divisor as E_α. So there are two base divisors with $\beta = 2$, not three. The redundancy remark in §4 covers β but not α.

(b) On $Z_{\emptyset,\alpha,\varepsilon}$ the form is $-x(\varepsilon)^5 z(\beta) z(\alpha)^{10} w(\varepsilon)^{-6}$, not $x(\varepsilon)^5 z(\beta) z(\alpha)^9 w(\delta)^{-6}$. So $b = 11$ and $\beta = 11/3$, not $10/3$, and the table entry $7/3$ should be $8/3$. $E_{\emptyset,\alpha,\varepsilon}$ and $E_{\emptyset,\alpha,\zeta}$ are one divisor.

(c) A divisor with $\beta = 11/3$ would contribute $\theta(\lambda)|\lambda|^{8/3}$, a term that cannot occur in the germ expansion. Its coefficient must therefore vanish, and the paper should show this, for example by a ramification argument for θ with $\theta^3 = \kappa_E$. The manuscript never mentions this divisor's contribution.

(d) The dismissal of exceptional divisors ("because of lemma 0.8") is circular, since Lemma 0.8 concerns strict transforms only. Z is singular along five torus-orbit closures (listed in the output of `g2_exceptional_divisors.py`). Any resolution adds rays u in the cone σ, and their Igusa constant is $\langle m_\omega, u\rangle / \langle \lambda, u\rangle$. Integral values cannot be excluded in advance. For example, $u = (3,2,2,2,3,5,7)$ lies in a singular face and has $\beta = 2$. Whether the $\beta = 2,3,4$ coefficients receive extra contributions therefore depends on the resolution chosen, and on the F-points and images of its exceptional divisors. The spurious divisors over $Y_1 \setminus Y''$ are likewise not treated for $r = 2,3$.

The proof of Lemma 5.14 also has gaps. The homogeneity degrees "by (1.22)" are asserted, not computed. Only the stratum $D = E \cap E'$ is considered, whereas Lemma 5.7 gives F_1 as a sum over strata. The contradiction requires the D-term to be nonzero. And $e = 1$ alone does not show that spurious divisors fail to contribute.

Verdict: *fixable gaps; the value $\beta = 10/3$ is false*. **Level:** L1 (scripts).

Issue 4.6 (The X-dependence and the measures on orbits and fibers)**. Where:** (6.5a), (6.5b), (7.3)–(7.5), (8.1)–(8.7). The decisive quantities $p_{\mathscr{O},T}(X)$ in (7.5) and (8.7) are asserted to be products over roots of H "using (1.22)". The exponents are never computed, and neither is the X-dependence of $\kappa(t_*)$. In addition:

- (7.3) uses $dz(\alpha)$, but $z(\alpha) = 0$ on $E_{\emptyset,\alpha,\delta}$. It lists six differentials for the 8-dimensional orbit $\mathscr{O}_2$, and it includes $dv(\alpha)$ although $\alpha \in \Sigma_{H_2}$. The Z-coordinates available on $\mathscr{O}_2$ are $w(\gamma)$, $w(\delta)$, $x(\delta)$ and $x(\varepsilon)$, and they should be paired with $v(\beta)$, $v(\gamma)$, $v(\delta)$ and $v(\varepsilon)$. "$\Sigma_{\mathscr{O}_3}$" in (7.3) should read $\Sigma_{\mathscr{O}_2}$.

- In (8.1) and (8.5), $w(\delta)$ should read $w(\zeta)$. With that correction, (8.1) agrees with my computation: $\omega_E = x(\zeta)^2 w(\zeta)^{-3} z(\alpha)^{-2}$. The display after (8.2) shows ω_Z, mislabeled as ω_E.
- For (8.4) I find $dx(\alpha)\,dx(\gamma)\,dx(\delta)/\lambda^2 = 2x(\zeta)w(\zeta)^{-1}z(\alpha)^{-2}\,dz(\alpha)\,dw(\gamma)\,dw(\delta)$, not $z(\alpha)^{-1}$. More fundamentally, defining a measure on $\mathscr{O}$ as a $\lambda \to 0$ limit of ratios is not justified. It is better to use the invariant measure on $\mathscr{O}$, which is unique up to a constant [R], and to compute $\omega_{E(u)}$ from the fibration.
- In §6, E_α yields the factor $|\beta(X)|$. For H_2, β is not a root of H. The untypeset text after §6 records an unresolved discrepancy of this kind.
- The translation invariance of $\kappa(t_*)$ "by continuity" needs a bound uniform in X, which is not given.
- Transfer needs one f^H for all (T,κ) attached to H. The theorems fix T_G before choosing f^H.

Verdict: *fixable gaps (substantial).* **Level:** L1.

Issue 4.7 (Lemma 3.5, Lemma 4.4 and the "regular branch")**. Where:** Lemma 3.5; §3 (the regular branch $E_\emptyset$); Lemma 4.4. The convention "Hales' E_S is my $E_{\Delta-S}$" has not been carried through. Three statements are affected.

- As printed, Lemma 3.5 gives for $S_1 = \Delta$ the generators $\lambda, x(\alpha), x(\beta), w(\gamma), \ldots, w(\zeta)$. Then $x(\gamma) = w(\gamma)x(\alpha)x(\beta)/\lambda$ is not regular. The correct generators are λ, $z(\alpha)$ for $\alpha \in \Delta - S_1$, $x(\alpha)$ for $\alpha \in S_1$, $x(\rho)$ for $\rho \notin \Sigma^+_{\Delta-S_1}$, and $w(\rho)$ for $\rho \in \Sigma^+_{\Delta-S_1} - \Delta$. The generator count then equals $p+1$.
- Lemma 4.4 should read $\beta(E_S) = 1 + |\Sigma^+_{\Delta-S}|$, with the parabolic $P_{\Delta-S}$. As printed, it gives $\beta(E_\Delta) = 7$, against the value 1 in the table.
- "The regular branch $E_\emptyset$ satisfies $\pi(E_\emptyset) = V$" is wrong in the paper's own convention. $E_\emptyset$ is the trivial-orbit branch ($\beta = 7$), and the regular branch is E_Δ.

The lists of generators in §4 are correct. I checked all twelve patches.

Verdict: *false as printed; misprint-level repair.* **Level:** L1.

Issue 4.8 (The definition of S'' and Y'')**. Where:** §0, Step 2; Definition 1.16. S'' is defined as "the image of S_0 in S_1" (and in Definition 1.16 as the image "in Y_1"). A point of S_0 is a regular star, so every $z(W,\rho) \neq 0$ there, and over $\lambda = 0$ the only element of $\bigcap_W B(W)$ that is unipotent is 1. The base divisors, on which $z(\alpha) = 0$, would then not lie in Y''. Presumably S'' is meant to be the open subset of S_1 on which all $z_1(W,\rho) \neq 0$, with $z(\alpha)$ arbitrary, as in Hales' thesis.

Verdict: *fixable gap.* **Level:** L1 for the inconsistency; L3 for the intended definition.

Issue 4.9 (Lemma 3.7 and the singular locus)**. Where:** Lemma 3.7; §3 (the claim that the singular locus lies in $K \cup K_0$); Conjecture 3.4. The proof of normality asserts saturation without checking it.

- For G_2 I proved saturation by an exhaustive exact computation, which covers every lattice point of the fundamental parallelepipeds (degree at most 6). So Z is normal for G_2. It is Cohen–Macaulay by Hochster's theorem [Ho], which is in the bibliography but never cited. For general G the claim is unproved.

- "Set-theoretic complete intersection" is not proved. The six equations (3.1) also vanish on the 10-dimensional set $\lambda = z(\alpha) = z(\beta) = 0$, so they do not cut out Z.
- The singular-locus claim is correct, but the fifth patch is redundant. On Z one has $x(\varepsilon)z(\alpha)^3 = x(\beta)w(\varepsilon)$, so that patch lies inside $\{x(\beta) \neq 0\}$.
- K contains the divisor $E_{\emptyset,\alpha,\delta}$, so "beginning with K as the center of the first blow-up" needs rewording.
- "The branches of $E_{Y''}$ are reduced" contradicts the multiplicities 2 and 3 found in §4.

Conjecture 3.4 does hold for G_2: all essential patches are regular, and every torus-invariant divisor of Z meets one of them. The text proves only the weaker statement about $K \cup K_0$.

Verdict: *fixable gaps; the claims are true for* G_2. **Level:** L1 (scripts).

Issue 4.10 (Section 1 foundations)**.** The following points concern §1.

- *The map ϕ is not well-defined.* A regular conjugacy class meets T_G in a full Weyl orbit, so "the point in T_G which meets the conjugacy class of g" is defined only in T_G/Ω. The star must be used to choose the point. (L1; fixable.)
- *Negation error.* Not lying in $S'(B_+, B_-)$ means that for *some* $\alpha \in \Delta$, $z = 0$ for *all* ω. The text says "for all α and ω" and concludes that the star lies on the diagonal. The intended conclusion, that such stars are not regular, still follows from Lemma 1.12. (L1; fixable.)
- $\rho = \omega\alpha$ (in the definition of a pair and of S') contradicts $\alpha_{i+1} = \omega_i\rho_{i+1}$; it should read $\rho = \omega^{-1}\alpha$. The sign in (1.8) is opposite to that in (1.9) and (1.10). In the definition of a root, $t^{-1}\theta_\rho(\xi)t$ should be $t\theta_\rho(\xi)t^{-1}$. "$\ker\rho \subset X^*(T) \otimes \mathbb{R}$" should be the hyperplane $\rho^\perp$. (L1; misprints.)
- *Equation (1.22).*
  - The c_1-term of $w(\gamma)$ lacks the factor $z(\sigma_\beta^{-1}W_0, \alpha)$ that appears in $x(\gamma)$.
  - The last term of $w(\zeta)$ lacks c_7 and has n_ζ for n_δ.
  - The constants $c_1, \ldots, c_7$ and the meaning of "higher order terms in λ" are never specified.
  - The final list gives seven "algebraically independent" coordinates. Together with λ that makes eight functions on the 7-dimensional patch $Y_1(B_+, B_-)$, since $z(W_0, \alpha) = 1/n_\alpha$ and $z(W_0, \beta) = 1/n_\beta$ already account for the star. The list needs correcting.

  (L1; fixable.)

Issue 4.11 (Section 2 computations)**.** The following points concern §2.

- The requirement $g^{-1}\sigma(g) \in T_G(\overline{F})$ for all σ would make B_G defined over F, which is impossible for elliptic T_G. It should be $g^{-1}\sigma(g) \in N(T_G)(\overline{F})$, and this gives σ_T. "Factors through $Gal(\overline{F}/E)$" should read "through $Gal(E/F)$". (L1.)
- In (2.8) and (2.9), b' should be $c_\sigma\sigma(n_{\sigma_T^{-1}W_0})\,\sigma(b)\,\sigma(n_{\sigma_T^{-1}W_0})^{-1}c_\sigma^{-1}$. The printed formula has b for $\sigma(b)$, c_σ in place of c_σ^{-1}, and a double inverse. (L1; misprint.)

- In $SL(2)$, $\theta_\alpha(x)^{\theta_{-\alpha}(z)} = \theta_{-\alpha}\big(\frac{-xz^2}{1+xz}\big)\,\alpha^\vee(1+xz)\,\theta_\alpha\big(\frac{x}{1+xz}\big)$. So $\phi_\alpha^\alpha = x(\alpha)/(1+x(\alpha)z)$, not $x(\alpha)$. The analogous y_β is $-xz^2/(1+xz)$. The torus factor also rescales the other coordinates. (L1.)
- In $\theta_\beta(r)\theta_\alpha(s)$ the exponents are swapped. The correct terms are $\theta_\delta(rs^2)\theta_\varepsilon(rs^3)\theta_\zeta(r^2s^3)$, because $i\alpha+j\beta$ carries s^ir^j. In ϕ_δ^α, c_2 should be c_1. In the sentence after Lemma 2.10, $\theta_\rho(z)$ should be $\theta_{-\rho}(z)$. In (2.17) the subscripts $\eta_{\alpha,\cdot}$ should be $\eta_{\beta,\cdot}$. (L1; misprints.)
- The figure of the Dynkin curve is missing: the source has empty displays.

Issue 4.12 (The appendix)**.** The following points concern the appendix.

- *A.2.*
  - $k \times j = i$ should be $k \times j = -i$.
  - The form $\frac{1}{2}(xy^* + y^*x)$ is not scalar-valued: for quaternions with $x = 1$ and $y = i$ it equals $-i$. The correct form is $\frac{1}{2}(xy^* + yx^*)$.
  - The triality theorem holds for proper orthogonal maps (rotations), not for arbitrary ones.
  - The "proof" checks only the infinitesimal identity, which does not imply the global statement. A reference such as Springer–Veldkamp would serve.

  (L1; L3 for the reference.)
- *A.4.* The matrix of $X_{\alpha+\beta}$ has entry -1 at position $(1,3)$, while the note says $E_{13}+E_{24}+E_{25}$. All other matrices check out (see Section 6). (L1.)
- *A.5.*
  - The displayed basis of P is wrong: $(1/2,0)\cdot\beta^\vee = -1/2$. The correct basis is $(1/2,\sqrt{3}/2)$, $(0,\sqrt{3})$. The conclusion $P = Q$ is right.
  - The $A_1 \times A_1$ subsystem should be generated by $\alpha^\vee$ and $\zeta^\vee$, not $\beta^\vee$ and $\zeta^\vee$, which are not orthogonal.
  - "$Q^\vee$ strictly contains $P^\vee$" should read "is strictly contained in $P_1^\vee$".
  - The argument identifying H_2 is hard to follow, though the answer is right: the character lattice Q is the index-2 sublattice of P_1 containing $(\alpha+\zeta)/2$.
  - For H_1, see Issue 4.3.

  (L1.)
- *A.7.* The stability arguments for $\mathscr{O}_2$ and $\mathscr{O}_3$ are plausible. I confirmed $\mathfrak{g}_2(2) = FX_{\alpha+\beta}$ and $\mathfrak{g}_2(2) = FX_\beta$ and the subregular grading. Two points need attention.
  - For $\mathscr{O}_2$ the grading is odd, so the use of Rao's lemma should be justified.
  - The claim "7 F-classes" in the stable subregular class conflicts with my count. The component group is $A(u) = S_3$ and $H^1(F, G_2) = 1$, so the classes correspond to étale cubic F-algebras. There are 8 of these when $q \equiv 1 \pmod 3$ and 6 when $q \equiv 2 \pmod 3$ ($p \neq 2,3$). Please check Assem's statement.

(L1 with L3 inputs.)

- *Untypeset weighted Dynkin diagrams.* The labels for the 2- and 3-regular classes are swapped. The long-root class A_1 (3-regular) has weights $(\alpha,\beta) = (0,1)$, and the short-root class $\tilde{A}_1$ (2-regular) has $(1,0)$. (L1.)

5 Minor issues, misprints and inconsistencies

All entries are L1 unless marked otherwise.

Location	Issue	Verdict
§0	"$T'_G \cap \{1\}$" should be $T'_G \cup \{1\}$.	misprint
§0	Property (e) sets $E = \pi^{-1}(u)$, a fiber. It should concern a divisor E with $\pi(E) = \overline{\mathcal{O}}$.	misprint
§0	The conjecture mixes $\Delta(\gamma_H, \gamma_\lambda)$ with $\Phi(\gamma_G, f)$.	misprint
§0	"apply lemma 5.18" should read "lemma 5.19". (5.18) is an equation.	misprint
§0	D_H is used without absolute values in (0.1) and in Formula 0.6.	misprint
§0	"Langland's" should be "Langlands's".	spelling
§0, §3	"Cohen-Macauley" should be "Cohen–Macaulay", here and in [Ho].	spelling
§0	Lemma 0.8 says it is "proven in §§4–5". The equality of the principal values over E and $\tilde{E}$ is never proved there.	fixable gap
§0	"desingularize Y'' by blowing up four times": the derivation is omitted, so the claim cannot be checked.	unverified
§1	"Lemma 1.2 below" does not exist. Tag (1.3) is missing.	misprint
§1	[Sp3] is cited but missing from the bibliography.	misprint
§1	Lemma 1.21: the formula for $c_{\rho_1,\ldots,\rho_m}$ is not a precise statement.	readability
§2	"the projection map π of §1.1": there is no §1.1. X^0 should be X_0.	misprint
§3	In g2new3 the macro `\off` means $\mathcal{O}_F$, so the first line prints "over $\mathcal{O}_F$". $\overline{F}$ was probably intended.	misprint
§3	$x(\beta) \neq 0$ patch: $w(\zeta) = x(\alpha)x(\zeta)z(\alpha)^4/x(\beta)^2$.	misprint
§3	$x(\gamma) \neq 0$ patch: the exponent of $x(\gamma)$ in $w(\zeta)$ is 4.	misprint
§3	$z(\alpha) \neq 0$ patch: $w(\zeta) = x(\zeta)w(\varepsilon)z(\beta)/x(\varepsilon)$, with no $z(\alpha)^{-3}$.	misprint
§3	Lemma 3.6: the statement has $\rho > \sigma$, the proof uses $\rho \geq \sigma$. In general G, roots incomparable with σ are not covered.	misprint / gap
§3	"known ... for C_n [H1], [H2]": [H2] treats only $GSp(4)$. "False for D_n" is unsupported.	L2/L3
§3	[Sp] is not a bibliography label.	misprint

Location	Issue	Verdict
§4	In $\omega_{Y''(B_+)}$ there is a doubled $\wedge$.	misprint
§4	$\sigma = \varepsilon, \zeta$ forms: the denominator should be $w(\varepsilon)$, $w(\zeta)$, not $w(\delta)$.	misprint
§4	Lemma 4.4: the second case of Ind should read "3-regular".	misprint
§4	Observation (II): for strict transforms, β is unchanged whether or not E contains a center, because Z is normal.	readability
§5	Lemma 5.10(c): the chain $\overline{\mathscr{O}} = \mathscr{O} \cup \mathscr{O}_m$ is garbled.	misprint
§5	In the proof of Lemma 5.7, $\sum_{1 \le i \le p} c_j/(j-1)!$ has the wrong index and a spurious $(j-1)!$.	misprint
§5	In the proof of Lemma 5.17, "(5.10)" refers to a lemma, and "ξ satisfies (5.2)" is unexplained.	fixable gap
§5	Lemma 5.19: with $\Gamma^{T,\kappa}$ normalized by Δ, the homogeneity exponent is $2(r - \lvert\Sigma_H^+\rvert)$, not $2r$.	misprint
§6	$E_{\emptyset,\alpha,\gamma} = E_\alpha$; see Issue 4.5.	false
§6	(6.5b) drops the prefactor $\lvert\lambda\rvert\kappa(\cdot)D_H^{-1}$.	misprint
§6	"E is E_α or E_β, depending on T_G" is unexplained, since both are $\beta = 2$ divisors.	fixable gap
§6	In g2new6 `\circ ff` means $\overline{F}$, so the text prints $\exp(\overline{F}X)$. $\mathscr{O}_F$ may have been intended.	misprint
§7	"$\Sigma_{\mathscr{O}_3}$" in (7.3) should read $\Sigma_{\mathscr{O}_2}$. The exponent $3/2$ of $\lvert\lambda\rvert$ is unexplained.	misprint
§7	F-rationality of $E_{\emptyset,\alpha,\delta}$ via observation (I) needs the uniqueness argument spelled out.	readability
§8	(8.6) should have $\lvert\lambda\rvert^3$, not $\lvert\lambda\rvert^2$; compare the display before (8.2).	misprint
§8	"conjecture 3.4 (which we proved in §3)": §3 does not prove it.	false (but true)
§9	Variant g2new9d: `\circ ff` means $\mathscr{O}_F$, so (5) and (6) print $G(\mathscr{O}_F)$, $H(\mathscr{O}_F)$. [A] and [H4] are not in the bibliography.	misprint
App. A.6	"(a) simple reflections $\langle s_\alpha, s_\zeta \rangle$": s_ζ is not simple.	misprint
Untypeset	The "Lemma 9.1" in the text after §6 belongs to no section. "Lemma 1.21" in the text after §2 duplicates Lemma 6.4.	housekeeping
Throughout	"homogeniety", "throughtout", "neighborhoood", "occuring", "patched" (for patches), "Automomorphic", "Represenattions", "Birkhauser".	spelling

6 Results I verified as correct

- *Normality.* Z is normal for G_2. The essential patches listed in §4 are all regular and cover Z up to codimension two, so Conjecture 3.4 holds for G_2. (L1.)
- *Igusa constants.* $\beta = 1$ ($E_{\{\alpha,\beta\}}$), 2 (E_α, E_β), 3 with $a = 2$, $b = 6$ ($E_{\emptyset,\alpha,\delta}$), 4 with $a = 2$, $b = 8$ ($E_{\emptyset,\beta,\zeta}$), and 7 (trivial orbit). The forms on Z_α, Z_β, $Z_{\emptyset,\alpha,\gamma}$, $Z_{\emptyset,\alpha,\delta}$ and $Z_{\emptyset,\alpha,\zeta}$ agree with mine up to the misprinted denominators. (L1.)
- *Patch formulas.* The patch formulas of §3 are correct apart from the three misprints listed. (L1.)
- *Orbit bookkeeping.* The orbit dimensions $12-2r$, the homogeneity of the normalized germs ($|\lambda|^{2r}$), the relation $\beta - 1 = r$, and the dimension counts $\dim \mathscr{O}_2 = 8 = 14 - \dim SO(4)$ and $\dim \mathscr{O}_3 = 6 = 14 - \dim SL(3)$ are correct. (L1.)
- *Lemma 5.6* is correct. (L1.)
- *Triality.* The root bookkeeping of A.3 and (A.11) is correct: the fixed and permuted roots of D_4 and the formulas for $\phi(\varepsilon_i)$. So are all Chevalley matrices X_σ, h_σ in A.4 apart from the sign noted, and the torus T' with $\alpha = a_3$, $\beta = a_2 a_3^{-1}$. (L1.)
- H_2. The endoscopic group $H_2 = (SL(2) \times SL(2))/\{\pm 1\}$ with $\Sigma^+_{H_2} = \{\alpha, \zeta\}$ is correct, and $H_1 \cong PGL(3)$ as an abstract group is correct. (L1.)
- *Rational orbits.* The weighted gradings in A.7 are right, as is the conclusion that $\mathscr{O}_2$ and $\mathscr{O}_3$ each consist of a single rational orbit. (L1 for the gradings.)

7 Readability

1. *Notation clashes.*
   - β is both the long simple root and the Igusa exponent b/a. For example, "$\beta(E_{\emptyset,\beta,\zeta}) = 4$".
   - γ is a root, a group element and a constant in (5.2).
   - δ is a root, a cohomology class, and the blow-up indicator in (II).
   - ε is a root and the Mellin cutoff. p is the prime, $|\Sigma^+|$ and an index in Lemma 5.7. π is the projection and the uniformizer.
   - σ is a Galois element, a root and the map $S_2 \to \Sigma^+$. α is also the unit in (5.1).
   - E is a divisor and the splitting field. S_1 is a subset of Δ and the variety S_1. K is a subvariety and a maximal compact subgroup.

   I suggest renaming the Igusa exponent (for example $\nu(E)$) and the roots $\gamma, \ldots, \zeta$ (for example $\rho_3, \ldots, \rho_6$).
2. The macro `\off` means $\overline{F}$ in some files and $\mathscr{O}_F$ in others. This produces at least four printed errors.

3. Conventions are never fixed in one place: $B^g = g^{-1}Bg$, the meaning of σ_T, which Borel contains the positive roots, the sign of z, and the Chevalley normalization. The reader has to infer them from formulas, some of which conflict (Issue 4.10).

4. The constants c_i, d_i, e, f_i are "computable" but never computed. The phrase "higher order terms in λ" is used for expressions whose other coordinates also depend on λ.

5. Several key steps are "left to the reader", "omitted" or "similar": Lemma 5.6, the case $e = 3$ in Lemma 5.14, the four-step blow-up, and the details of (1.22).

6. The Dynkin curve figure is missing.

8 Bibliography check

Label	Finding	Level
[Ro]	Published as J. Rogawski, "An application of the building to orbital integrals", Compositio Math. 42 (1980) 417–423. The thesis title in the manuscript differs.	L2 (EuDML, Numdam)
[S2]	The title is "A formula for regular unipotent germs", Astérisque 171–172 (1989) 275–277.	L2 (Numdam)
[H2]	The pages are 195–256.	L2 (Numdam)
[H3]	The pages are 303–333 (Univ. Montréal, 1992).	L2 (AMS reference list)
[L]	The pages are 465–506.	L2 (IAS archive)
[LS0]	LNM 1041, pp. 250–279. The volume is dated 1984 (copyright 1983).	L2
[LS3]	Correct as listed.	L2
[K]	LNM 339 is dated 1973. The authors are Kempf, Knudsen, Mumford and Saint-Donat.	L3
[R]	The pages should be checked: I recall 505–510.	L3
[SpSt]	LNM 131 is from 1970. "1986 (2nd ed.)" should be checked.	L3
[Har]	The year (1977) is missing.	L3
[Rep]	This is part II (subregular term). Shelstad [S2] cites part I, PJM 113 (1984) 165–172.	L2/L3
missing	[Sp3], [Sp], [A], [H4], [D] are cited but have no entry.	L1
uncited	[Ho], [Rep′], [Sp1] are never cited. [Ho] is the natural reference for Lemma 3.7.	L1

9 Compliance with the house style guide

The manuscript is a 1994 draft, so these items describe what a revision under §§1–6 of the style guide will need. They are not criticisms of the original.

- *§2 (structure).* The paper has no abstract, AI-disclosure sentence, literature review, background section with examples, notation table, summary of results or glossary.
- *§1 (typesetting).* The source is AMS-TeX split over separate files with hand-typed numbers and hand-typed cross-references. A revision needs one amsart file, `\label`/`\ref`, theorem numbering within sections, and citation keys of the form `Ha89`. The transcription delivered with this report takes the first steps (single file, house-style keys, 12pt Times, letter paper, a clean pdflatex build) but keeps the hand-typed numbering on purpose.
- *§4 (status of results).* Theorems 6.1, 7.2 and 8.3 should be labeled conditional or conjectural until Issues 4.1–4.6 are resolved. The same applies to Lemma 3.7 for general G, to Lemma 0.8, and to Lemma 5.14.
- *§5 (authorship).* The author line should give `wdjoyner@gmail.com` in place of the USNA address. The acknowledgments need the AI-usage statement.
- *§6 (revisions).* The title, the subtitle "Draft *n* — Corrected and expanded version" and the dates need to be added. A revision log is needed, with a list of "Remark 2026.*n*" items for unverified claims.

10 Notes on the transcription

The single-file transcription follows §6.7 of the style guide: it is a literal transcription with no corrections. All errors listed above are kept. Additions by me appear only as comments beginning "Claude:". Specific choices:

1. The body follows `g2new5.tex` and `g2new9.tex`. Every difference in the variants `g2new5d.tex` and `g2new9d.tex` is transcribed in a comment or an `\iffalse` block marked "VARIANT". I could not tell which version is later. The "d" files align §5 with the wording of §8 and extend Conjecture 9.1. They also cite [A] and [H4], which are missing from the bibliography.
2. The text after `\enddocument` is transcribed in `\iffalse` blocks marked "UNTYPESET". All of these blocks compile when switched to `\iftrue`.
3. The macro `\off` is rendered as each file printed it, and doubtful cases are flagged.
4. Missing citation labels are typeset literally.
5. Underlined bibliography titles are set in italics.
6. Long inline formulas were allowed to break after commas. This avoids overfull boxes and changes no content.

11 Recommendation

Major revision. The geometric core is worth preserving, and with the corrections above it is largely right. A revised version should do five things.

1. Replace the appeal to Lemma 5.19 by an explicit computation of the X-dependence, including the κ-factor, in the style of [H2, §6].
2. Restate the theorems as germ identities, or with orbital-integral conditions on f.
3. Use the correct root system for $PGL(3)$, and treat the quadratic and S_3 tori.
4. Account for the divisor with $\beta = 11/3$ and for the exceptional and spurious divisors.
5. Fix the misprints and the notation clashes.

Until then, the main results should carry the label "conditional" (style guide §4.1).

Acknowledgments and AI disclosure

This report was written by Claude (Opus 5.5, Anthropic) acting as referee at the request of the author. The computations were run by Claude in Python with SymPy and SciPy, and the scripts accompany the report. AI review is not an independent certification of the results, and statements at level L3 should be checked against the sources. Further details on the AI usage belong in the revision log.